

Operation Certainty

DamageBDD × ECAI — Deterministic Verification for Enterprise

A phone-friendly, one-page executive brief

One-liner

Operation Certainty turns Behavior-Driven Development (BDD) into a *deterministic, audit-ready revenue engine*: tests become on-chain verification artifacts; ECAI encodes knowledge as elliptic-curve state for zero-guess retrieval; results trigger automated payouts and compliance proofs.

The Problem & the Prize

- Software failure, audit gaps, and AI hallucinations create material risk and cost in regulated industries (downtime, rework, fines, reputational loss).
- Tooling focuses on logs and dashboards, not *mathematically provable* behavior; leadership cannot “buy certainty.”
- **The Prize:** A verification / assurance market counted in the tens of billions (DevSecOps, QA, GRC). We propose building a **new ARR pillar** via deterministic verification subscriptions and milestone-based automation.¹

Our Left-Field Solution

- **DamageBDD:** Human-readable BDD specs compile to on-chain proofs (Aeternity/Sophia) and tamper-evident logs (IPFS/Arweave).
- **ECAI:** Elliptic-Curve AI encodes knowledge as curve points ("knowledge NFTs"); retrieval is deterministic, not probabilistic.
- **Automated Economics:** Lightning escrow for milestone completion; tokens (AEX9) meter verification and pay operators only when proofs pass.

Key Financials (illustrative)

Estimated Upfront Investment (12-mo pilot)	US\$1.2M
Projected 3-Year Revenue	USD US\$15–US\$25M (subscription + usage fees)
Projected ROI / IRR	8–12x / attractive IRR based on staged rollouts
Time to Break-Even	18–24 months (assumes 3 enterprise logos in Y1)

The Ask

- **Approve a 90-day, US\$200k** feasibility study (security, legal, pilot architecture).
- **Book a 30-minute** slot with the Strategic Committee next month for a live demo and pilot gating.
- Nominate an executive sponsor and 2 domain leads (Engineering and Risk/Compliance).

Principle: Sell the next meeting, not the entire program.

¹Replace bracketed figures with company-validated estimates before distribution.

Operation Certainty — Supporting Document (3 pages max)

This appendix anticipates first-round questions and shows strategic fit.

1. Opportunity & Strategic Fit

Market & Customer

Who: regulated and reliability-critical sectors (financial services, healthcare, critical infrastructure, gov/defense, enterprise SaaS). Their pain: provable behavior, auditability, & reduction of AI risk.

	TAM	SAM	SOM (Y3)
Deterministic Verification (QA/DevSecOps/GRC)	[US\$ XX B]	[US\$ X B]	[US\$ M]

Competitive Landscape

Category	Positioning vs. Operation Certainty
Legacy QA/CI tools	Great at pipelines; weak at cryptographic proofs and pay-on-proof economics.
AI testing platforms	Useful insights; probabilistic and non-auditable decisions create residual risk.
GRC suites	Strong workflows; lack executable, machine-verifiable behavior contracts.
DamageBDD × ECAI	Deterministic proofs, economic finality, and audit-grade artifacts by design.

Strategic Alignment

- **Risk Reduction:** Cryptographic proofs replace faith-based attestations; lowers audit and model risk.
- **New ARR Pillar:** Verification-as-a-Service (VaaS) subscriptions + usage-based fees.
- **Differentiation:** PROVABLE software behavior; defensible moat via protocol, proofs, and network effects (operators/validators).

2. Solution & Business Model

Solution Blueprint (high level)

1. **Author:** Teams write BDD scenarios in plain language (Given/When/Then) with compliance tags.
2. **Encode:** ECAI maps knowledge to elliptic-curve points (knowledge NFTs); BDD compiles to Sophia contracts.
3. **Verify:** Nodes execute steps; passing proofs are anchored on-chain and content-addressed (IPFS/Arweave).

4. **Settle:** Lightning escrow releases payouts on proof; failures pay nothing, creating *pay-on-proof* economics.
5. **Report:** Auditors and stakeholders consume immutable verification trails and dashboards.

Business Model

Tier	What you get	Indicative Price
Starter (Team)	1 project, capped scenarios, hosted verification	[US\$ 2k/mo]
Enterprise	Unlimited projects, private registry, SSO, SLAs	[US\$ 25k+/mo]
Regulated/On-Prem	Air-gapped nodes, HSM integration, custom reporting	[Quote]
Usage Fees	Per-proof metering (AEX9 tokenized), marketplace operator rewards	[Per proof]

Go-to-Market (first 10, then 100 customers)

- **Land:** Target 2–3 lighthouse use-cases with high audit pressure (payments, trading, clinical, critical infra).
- **Expand:** Show measurable % fewer defects and audit hours; templatize playbooks; enable partner operators.
- **Scale:** Certify external nodes; offer shared-revenue catalogs of verified behaviors.

3. Risk, Team, & Next Steps

High-Level Financials (illustrative)

US\$M	Y1	Y2	Y3	Y4	Y5
Revenue	2.0	6.0	15.0	24.0	35.0
COGS	0.8	1.8	3.5	5.0	7.0
Gross Margin	1.2	4.2	11.5	19.0	28.0
OpEx (incl. R&D, GTM)	2.2	3.2	5.0	6.6	8.0
EBITDA	-1.0	1.0	6.5	12.4	20.0

Risks & Mitigations

Risk	Mitigation
Market adoption	Start with regulated, high-pain domains; measurable ROI; executive champion.
Technology maturity	Stage pilots; formal verification on critical paths; independent security review.
Compliance/Legal	Early engagement with legal/audit; data residency options; on-prem variant.
Token / payments optics	Abstract token UX; clear accounting; use Lightning for fiat on/off-ramps; no speculation.

Team & Resources (pilot)

- 1 Program Lead, 2 Senior Engineers (Erlang/Sophia), 1 DevOps/SRE, 1 Security/Compliance lead, 1 UX.
- Budget covers infra, audits, HSM/PIV integration, pilot support, and partner enablement.

Roadmap & Milestones (first 12–18 months)

Quarter	Milestones
Q1 (Feasibility)	Threat model; legal review; pilot architecture; first BDD-to-chain demo; operator onboarding.
Q2 (Pilot Build)	Two lighthouse use-cases live; automated payouts; audit dashboards; SOC2 readiness plan.
Q3 (Scale-Up)	Private registry; partner node certification; enterprise SSO; tiered pricing in market.
Q4 (Expand)	Multi-LOB rollout; compliance packs (PCI/HIPAA/ISO); marketplace for verified behaviors.

Decision & Next Steps

- Approve the **90-day feasibility** and **schedule the committee demo**.
- Identify 2 high-pain use-cases and name an exec sponsor.
- Confirm security review scope (incl. key management, on-prem option).

Notes: All financials are illustrative placeholders pending due diligence. Replace bracketed figures with validated values before external circulation.